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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

EC3222-1 – EMBEDDED SYSTEMS

(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Design metrics in embedded system design are used to measure:
A) Cost, performance, power, and reliability B) Only the speed of the system
C) Software development time only D) None of the above
2. In embedded systems, FPGA, ASIC, and VLSI are examples of:
A) Processor technologies B) IC design technologies
C) Debugging tools D) Programming languages.
3. The monetary cost of manufacturing each copy of the system is called
A) NRE cost B) Unit cost
C) Total cost D) None of the above
4. The ability to change the functionality of the system without incurring heavy NRE cost is called
A) Correctness B) Unit cost
C) Performance D) Flexibility
5. Which of the following correctly represents the top-down design process in embedded systems?
A) Start from low-level circuit → move to high-level requirements B) Start from high-level specification → refine into architecture → detailed design → implementation
C) Begin with coding → then test → then design specification D) No sequence is required
6. Which of the following types of memory is typically used for permanent storage of embedded firmware?
A) RAM B) Flash/ROM
C) Cache D) Register file
7. Which is the storage element in DRAM?
A) Inductor B) Resistor
C) Capacitor D) MOSFET
8. What type of memory is suitable for high volume production?
A) RAM B) ROM
C) EPROM D) EEPROM
9. Which of the following is an application-specific embedded system example?
A) Laptop computer B) Washing machine controller
C) Tablet PC D) Gaming desktop
10. Which of the following is NOT an electronic control unit (ECU) in automobiles?
A) Engine Control Unit B) Airbag Control Unit
C) Anti-lock Braking System (ABS) Unit D) Text Editor Unit

11. The primary goal of hardware-software co-design is to:
- A) Maximize the hardware components and minimize software. B) Minimize the hardware components and maximize software.
- C) Achieve an optimal balance between hardware and software to meet system constraints. D) Design hardware and software completely independently.
12. The process of deciding which system functionality to implement in hardware and which in software is known as:
- A) High-level transformation. B) Hardware/software partitioning.
- C) Algorithmic-level design. D) Interfacing.
13. UML stands for:
- A) Unified Modeling Language B) Universal Machine Logic
- C) User Memory Layout D) Unit Model Language
14. In firmware design, a super loop approach is characterized by:
- A) Using an infinite loop to handle tasks sequentially B) Using a multitasking OS
- C) Running multiple processes in parallel D) Avoiding loops in the program
15. Interaction Diagram is a combined term for
- A) Sequence Diagram + Collaboration Diagram B) Activity Diagram + State Chart Diagram
- C) Deployment Diagram + Collaboration Diagram D) None of the mentioned
16. Which of the following diagram is time oriented?
- A) Collaboration B) Sequence
- C) Activity D) None of the mentioned
17. The main purpose of a cross-compiler is to:
- A) Translate assembly to binary B) Compile code for a different architecture
- C) Test code without hardware D) Convert machine code back to assembly
18. Which of the following integrates editor, compiler, linker, and debugger in one tool?
- A) IDE B) Assembler
- C) Emulator D) Loader
19. Which architecture in digital signal processor reduces the execution time?
- A) Harvard B) CISC
- C) Program storage D) Von Neumann
20. What does API stand for?
- A) address programming interface B) application programming interface
- C) accessing peripheral through interface D) address programming interface

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

		Marks	BT*	CO *	PO*
1.	a) Mention the significance of an embedded system. Give examples for the same and explain digital camera example.	8	L*2	1	1
	b) Explain the different external communication interfaces in brief.	8	L2	2	1
2.	a) Explain the roles of Processor, IC and Design technologies and explain the context in which they are used.	8	L2	1	1
	b) Describe a simple algorithm for finding the GCD of two integer numbers. Bring out the controller and data path for this algorithm.	8	L3	1	1
3.	a) What is the function of an actuator? Explain its role in embedded system design? Illustrate with example.	8	L2	2	1
	b) Design a 3-bit counter that counts the following sequence: 1, 2, 4, 5, 7, 1, 2, etc. This counter has an output "odd" whose value is 1 when the current count value is odd. Use the sequential design technique. Start from a state diagram, draw the state table, minimize the logic, and draw the final circuit.	8	L3	1	1

Unit – II

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|----|----|--|---|----|---|---|
| 4. | a) | Explain the different electronic control units and communication buses used in automotive systems. | 8 | L2 | 3 | 1 |
| | b) | What is the difference between Super loop based and OS based embedded firmware design? Which one is the better approach? | 8 | L2 | 4 | 1 |
| 5. | a) | Explain the sequential program model with an example of seat belt warning system. | 8 | L2 | 3 | 1 |
| | b) | Explain the different types of UML things and their significance in each stage of the system development life cycle. | 8 | L2 | 4 | 1 |
| 6. | a) | Explain the process of conversion from assembly language to machine language with necessary sketches. | 8 | L2 | 4 | 1 |
| | b) | What are the commonly used computational models in embedded design? Explain DFG and CDFG models in detail with example. | 8 | L3 | 3 | 1 |

Unit – III

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|----|----|--|---|----|---|---|
| 7. | a) | List and explain the various hardware debugging tools used in embedded product development. | 8 | L2 | 5 | 1 |
| | b) | Mention the difference between an assembler and a disassembler? State their use in embedded application development. | 8 | L2 | 5 | 1 |
| 8. | a) | Explain the advantages and limitations of simulator based debugging. | 8 | L2 | 5 | 1 |
| | b) | What is cross compilation? Explain in detail the need for cross compilation. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
